INCH-POUND
MIL-DTL-24643/21G
27 November 2023
SUPERSEDING
MIL-DTL-24643/21F
1 October 2009

DETAIL SPECIFICATION SHEET

# CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 300 VOLTS, TYPES LSTCJU AND LSTCTU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

| | | |
|---|---|---|
| First | - | One iron and one constantan conductor in type LSTCTU, for type J thermocouple in accordance with ASTM E230/E230M; one copper and one constantan conductor in type LSTCTU, for type T thermocouple in accordance with ASTM E230/E230M (see table I for diameter). |
| Second | - | Thermoset insulation (see table I for diameter). |
| Third | - | An optional glass braid. |
| Fourth | - | Covering. Printed identification as follows: Type LSTCJU shall have the legend “8 (GRAY)” printed on the iron conductor and the legend “3 (RED)” printed on the constantan conductor. Type LSTCTU shall have the legend “6 (BLUE)” printed on the iron conductor and the legend “3 (RED)” printed on the constantan conductor (see table I for diameter). Outer surface to be printed shall appear black. |
| Fifth | - | The two conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly. |
| Sixth | - | An optional binder. |
| Seventh | - | Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.040 inch. See table I for cable overall diameter. |

NOTE: Manufacturer’s identification tape is required (see 3.4.9 of MIL-DTL-24643).



Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

| Military part no. M24643/21 | Type and size | Numbers of strands and strand diameter (inch) | Diameter over insulation min. avg. (inch) | Diameter over covering max. (inch) | Overall diameter (inch) | |
|---|---|---|---|---|---|---|
| | | | | | min. | max. |
| -01UN | LSTCJU-4 | 7/0.025 | 0.105 | 0.140 | 0.400 | 0.430 |
| -02UN | LSTCTU-4 | 7/0.025 | 0.105 | 0.140 | 0.400 | 0.430 |

REQUIREMENTS:

Qualification required.

INSPECTIONS:

**Basic electricals:**

| | |
|---|---|
| Voltage withstand (volts, root mean square, min.) | |
| Conductor to conductor | 3,000 |
| Insulation resistance (megohms/1,000 feet, min.) | |
| Conductor to conductor | 500 |
| ASTM E230/E230M calibration limits of error (°F [°C]) | |
| Type LSTCJU | ±3.96 (±2.2) |
| Type LSTCTU | ±31.8 (±1) |
| Conductor continuity | No failure |

**Group A:**

| | |
|---|---|
| Visual and dimensional | No failure |
| Abrasion resistance (extruded insulation only) (scrapes, min.) | 250 |
| Watertightness (see MIL-DTL-24643 for limits of water leakage) | No failure |
| Crack resistance (extruded insulation only) | No damage |

**Group B:**

| | |
|---|---|
| Cold bending (cable) | No failure |
| Thermoset proof test (percent, max.) | |
| Insulation | 50 |
| Jacket (when tested at 392 °F [200 °C]) | 50 |
| Drip (203±1.8 °F [95±1 °C]) | No failure |
| Physicals (unaged) | |
| Insulation (extruded insulation only) | |



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| | |
|---|---|
| Tensile strength (lb/in$^2$, min.) | 700 |
| Elongation (percent, min.) | 150 |
| Jacket (cable) | |
| Tensile strength (lb/in$^2$, min.) | 1,300 |
| Elongation (percent, min.) | 160 |
| Tear (lb/in thickness, min.) | 35 |

**Group C:**

| | |
|---|---|
| Physicals (aged) (air oven) | |
| Jacket (cable) | |
| Tensile strength (percent of unaged, min.) | 60 |
| Elongation (percent of unaged, min.) | 60 |
| Permanence of printing (insulation) (cycles, min.) | 25 |
| Permanence of printing (jacket) (cycles, min.) | 125 |
| Heat distortion (percent of unaged, max.) | 30 |
| Cable fill (sealant) removability | No failure |
| Shrinkage | No failure |

**Group D:**

| | |
|---|---|
| Gas flame (circuit integrity) (1 hour) | No failure |
| Flame propagation (cable) | No failure |

**Qualification inspection:**

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

| | |
|---|---|
| Cold working (-4±3.6 °F [-20±2 °C]) | No damage |
| Gas flame (circuit integrity) (3 hours) | No failure |
| Aging and compatibility (cable) (257±9 °F [125±5 °C]) | No failure |
| Abrasion resistance (jacket) (scrapes, min.) | 75 |
| Acid gas equivalent (percent, max.) | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content (percent, max.) | |
| Jacket | 0.2 |
| Fillers | 0.2 |



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| | |
|---|---|
| Insulation | 0.2 |
| Immersion (jacket) | |
| Tensile strength (percent of unaged, min.) | 50 |
| Elongation (percent of unaged, min.) | 50 |
| Smoke index (max.) | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index (max.) | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) – type A (hardness, min.) | 80 |
| Weathering (jacket) | No failure |
| Electrical moisture absorption | No failure |

UNIT ORDERING LENGTHS:

| Type and size | Feet (nominal) |
|---|---|
| LSTCJU-4 and LSTCTU-4 | 2,000 |

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-037)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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